

Bhavik Patel

(832) 293-7191 | bpatel2001h@gmail.com | [linkedin.com/in/bpatel2001](https://www.linkedin.com/in/bpatel2001) | github.com/bpatel2001 | bpatel2001.github.io

EDUCATION

University of Houston

Bachelor of Science in Computer Science (Honors: Dean's List)

August 2024

Houston, TX

EXPERIENCE

Software Engineer Intern

June 2021 – September 2021

Sycamore Life Sciences

Houston, TX

- Developed a full-stack company website using HTML, CSS and JavaScript, integrating a MySQL-backed infrastructure with RESTful API to update, fetch and render product data, improving page load times by 40%.
- Designed UI components with JavaScript and CSS, using client-side rendering techniques such as asynchronous data fetching via AJAX/Fetch API and DOM manipulation, ensuring an interactive user experience.
- Implemented client-side caching using localStorage and sessionStorage, improving page load speeds by efficiently storing frequently accessed data locally, minimizing latency and reducing redundant API calls by 30%.
- Automated pricing analysis for 250+ products using Python, Selenium, and BeautifulSoup by aggregating and filtering market data, reducing manual research time by 80% and enabling data-driven pricing decisions.
- Built a data visualization dashboard using NumPy and matplotlib, allowing for adjustments based on market.

PROJECTS

Global Food Safety Website | *Java, Spring Boot, React, TypeScript, MySQL*

January 2025

- Developed a full-stack website with Java and Spring Boot to display global food statistics for 100+ countries.
- Built front-end using React, HTML, CSS, and TypeScript, providing responsive visuals and UI components.
- Established a RESTful API using MySQL and Hibernate, supplying CRUD endpoints for seamless data retrieval.
- Implemented front-end and back-end caching strategies, reducing server workload and website load times by 30%.

Gas Price Estimation Website | *MongoDB, Express, JavaScript, Node.js*

November 2024

- Delivered a full-stack gas price estimation website with HTML, CSS and JavaScript within an Agile SDLC.
- Secured user authentication using JSON Web Token (JWT) and OAuth2 to manage access control effectively.
- Created a RESTful API using Node.js and Express, enabling seamless data transfer and backend communication.
- Applied a NoSQL database schema with MongoDB, achieving flexibility and scalability for CRUD operations.

Chip-8 Interpreter | *Python, Pygame, Docker*

August 2024

- Built a Chip-8 virtual machine using Python, implementing 30+ functions to execute and run classic programs.
- Optimized CPU timing using multithreading, ensuring 700-1000 instructions/second with 60 Hz delay timers.
- Containerized program with Docker, enabling CI/CD by creating portable and reproducible environments.
- Employed a test suite to validate the accuracy of emulation, achieving 97% accuracy to ensure high performance.

Video Game | *C#, Unity, GitLab*

January 2024

- Conducted the core development of *Back To Life* – A 2D stealth-action game, in Unity, alongside a team of 7.
- Used C# and Unity tools to enable dynamic lighting using URP, AI pathfinding, and interactable objects.
- Enabled CI/CD pipelines using GitLab in order to automate testing, building, and deployment in collaboration.
- Presented to a panel of 10 industry professionals, gaining invaluable insight into strengths and shortcomings.

EXTRACURRICULARS

CougarCS

January 2022 – July 2024

- Led hands-on hardware and programming workshops, teaching 50+ attendees about embedded systems, starting with soldering and hardware assembly, showing participants proper soldering techniques and troubleshooting.
- Introduced firmware development in C, guiding members through compiling their code for micro-controllers using QMK, along with flashing the firmware and verifying its integrity through testing methods.

TECHNICAL SKILLS

Languages: Java, Python, C#, JavaScript, TypeScript, SQL, HTML, CSS, R

Frameworks/Libraries: Node.js, Express.js, Hibernate, Tailwind CSS, React, NumPy, Selenium, BeautifulSoup

Developer Tools: Git, Linux, MongoDB, MySQL, Gradle, Magento, npm, Vite, Posit, Docker

Relevant Courses: Data Structures/Algorithms, Database Systems, Software Design, Computer Architecture